Supplementary Materials for: Online Traveling Salesman Games: Core Fragility and the Temporal Nucleolus under Dynamic Arrivals

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Electronic Companion

Overview

This document provides supplementary material for the above manuscript. §S1 breaks down the aggregate Core-existence rates of the main manuscript's Section 6.4 by arrival pattern and coalition size. §S2 documents the empirical verification of the dimensionless joint-rescaling invariance declared in the main manuscript's Section 6.1. §S3 details the restricted Core LP methodology used for scale-up instances where $2^n - 1$ enumeration is infeasible, together with the full binding-size distributions for the 10 near-complement and 9 intermediate-coalition main-grid cases under nearest-neighbor (NN) dispatch, and includes a note on the Temporal Nucleolus cascade implementation. §S4 reports the empirical coverage audit of the balanced-near-complement threshold (Corollary 13) on the 10 near-complement main-grid instances. §S5 reports the per-policy robustness analysis under nearest-neighbor (NN), cheapest-insertion (CI), and batch re-optimization (BR) dispatch over the 175-instance grid in full detail, including Table S5 and the empty-mechanism breakdown summarized in Section 6.6 of the main manuscript. §S6 works out the Solomon C101 illustrative example, states and proves the reduction of the Online TSG to the static game at simultaneous arrivals, and reports its numerical verification on 25 pattern-A instances. §S7 states and proves the inheritance of static Core stability under exact online dispatch. §S8 records two auxiliary feasibility lemmas (Lemmas 4 and 5) lifted from the main manuscript for compactness. §S9 reports the partition-pair certificates for the 9 intermediate-coalition cases of Observation 15, supporting Theorem 14. §S10 contains the threshold-distribution histograms for r^{**}, r^{***}, and r^{*}.

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Cross-document references to Theorems, Propositions, Corollaries, Remarks, Observations, Sections, Figures, and Tables resolve via the `xr` package: numbers without prefix refer to the main manuscript; supplementary objects are cited as "Supplementary Proposition Sx," etc., or by section anchor.

S1 Per-pattern Core-existence breakdown

Figure S1 breaks down the aggregate Core-existence rates of Section 6.4 of the main manuscript by arrival pattern and coalition size n under nearest-neighbor (NN) dispatch. Two patterns retain 100% Core existence across all $n \in \{5, 7, 10, 12, 15\}$: light-load sequential $\mathrm{B}_{\mathrm{light}}$ ($\rho = 0.5$) and reverse-zig-zag D ($\rho = 1$). Pattern E ($\rho = 1$, random) holds at 100% for $n \in \{5, 7, 15\}$ and dips to 80% at $n \in \{10, 12\}$ via the intermediate-coalition mechanism of Observation 15. Pattern $\mathrm{B}_{\mathrm{medium}}$ fluctuates between 40% and 80% ($n = 5$:80%, $n = 7$:40%, $n = 10$:40%, $n = 12$:40%, $n = 15$:80%), again driven by intermediate-coalition empties at $k < n-1$. Heavy-load $\mathrm{B}_{\mathrm{heavy}}$ ($\rho = 4$) and clustered C ($\rho = 2$) are Core-fragile across the grid: $\mathrm{B}_{\mathrm{heavy}}$ shows 60%, 20%, 0%, 0%, 20% as n increases, while C shows 40%, 60%, 20%, 20%, 0%. Simultaneous arrivals (A) collapse non-monotonically from 60% at $n \in \{5, 7\}$ to 20%, 40%, and finally 0% at $n = 15$. Figure 2 of the main manuscript summarizes this by peak-queue regime k, showing how $k = n$ (saturated) drives most empty Cores while $k < n-1$ (shallow queue) preserves Core existence except for the intermediate-coalition cases in $\mathrm{B}_{\mathrm{medium}}$ and E.

S2 Scale invariance: empirical verification

This section verifies empirically the scale invariance stated in Section 6.1 of the main manuscript: that all dimensionless observables are invariant under the joint rescaling $(L, v, \tau) \to (\alpha L, \alpha v, \alpha\tau)$ for any $\alpha > 0$. Since we fix $v = 1$ throughout, this reduces to the simultaneous scaling $(L, \tau) \to (\alpha L, \alpha\tau)$ with the ratio L/τ (hence the load parameter ρ) preserved.

Experimental setup. Fixing seed = 42, we generate instances at $n \in \{10, 20\}$ and patterns $\{\mathrm{B}_{\mathrm{heavy}}, \mathrm{B}_{\mathrm{medium}}, C\}$ under four scale factors $\alpha \in \{0.5, 1.0, 2.0, 5.0\}$. For each configuration, coordinates are sampled from $[0, \alpha\sqrt{n}]^2$ and arrival intervals are scaled from the pattern-prescribed base $\tau_0 = \beta_{\mathrm{BHH}}/\rho$ to $\alpha\tau_0$. The nearest-neighbor dispatch policy runs on all 24 instances. Statistics r, r^{**}, and k are recorded.

Results. Table S1 reports the full 24 instances. For each of the six $(n, \mathrm{pattern})$ configurations, all four α values yield *bit-identical* statistics: the relative standard deviation of r across α is 0 to machine precision, and k and r^{**} are unchanged across α in every configuration. Only the dimensional quantities L and τ scale linearly with α as expected.

Table S1: Scale invariance verification (NN policy, seed = 42). Each row corresponds to one $(n, \text{pattern}, \alpha)$ configuration; 24 rows total. Within each $(n, \text{pattern})$ block, the four α values yield identical r, r^{**}, and k, confirming joint-rescaling invariance to machine precision. Em-dashes in r^{**} indicate configurations where the single-complement threshold of the main manuscript's Corollary 9 is inapplicable (no complement coalition feasible).

n	Pattern	α	L	τ	r	r^{**}	k
10	B_{heavy}	0.5	1.5811	0.0891	1.4225	1.1324	10
10	B_{heavy}	1.0	3.1623	0.1781	1.4225	1.1324	10
10	B_{heavy}	2.0	6.3246	0.3562	1.4225	1.1324	10
10	B_{heavy}	5.0	15.8114	0.8905	1.4225	1.1324	10
10	B_{medium}	0.5	1.5811	0.1781	1.4225	1.1324	10
10	B_{medium}	1.0	3.1623	0.3562	1.4225	1.1324	10
10	B_{medium}	2.0	6.3246	0.7124	1.4225	1.1324	10
10	B_{medium}	5.0	15.8114	1.7810	1.4225	1.1324	10
10	C	0.5	1.5811	0.1781	1.2531	1.1324	10
10	C	1.0	3.1623	0.3562	1.2531	1.1324	10
10	C	2.0	6.3246	0.7124	1.2531	1.1324	10
10	C	5.0	15.8114	1.7810	1.2531	1.1324	10
20	B_{heavy}	0.5	2.2361	0.0891	1.4896	1.1039	20
20	B_{heavy}	1.0	4.4721	0.1781	1.4896	1.1039	20
20	B_{heavy}	2.0	8.9443	0.3562	1.4896	1.1039	20
20	B_{heavy}	5.0	22.3607	0.8905	1.4896	1.1039	20
20	B_{medium}	0.5	2.2361	0.1781	1.4896	—	17
20	B_{medium}	1.0	4.4721	0.3562	1.4896	—	17
20	B_{medium}	2.0	8.9443	0.7124	1.4896	—	17
20	B_{medium}	5.0	22.3607	1.7810	1.4896	—	17
20	C	0.5	2.2361	0.1781	1.3469	1.1039	20
20	C	1.0	4.4721	0.3562	1.3469	1.1039	20
20	C	2.0	8.9443	0.7124	1.3469	1.1039	20
20	C	5.0	22.3607	1.7810	1.3469	1.1039	20

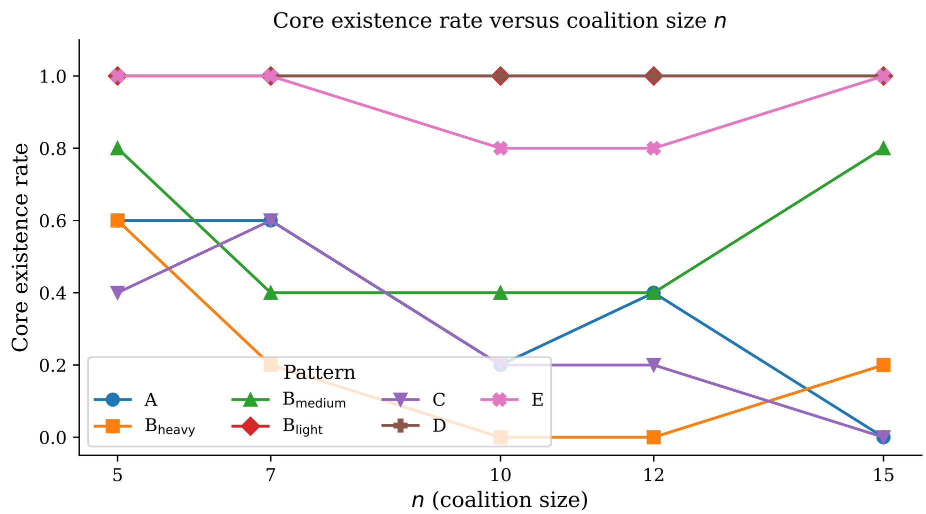

Figure S1: Core-existence rate versus coalition size n, broken down by arrival pattern. Patterns B_{light} and D overlap at 1.00 across the grid. Pattern E holds at 1.00 except at $n \in \{10, 12\}$ where it dips to 0.80. B_{medium} fluctuates between 0.40 and 0.80. B_{heavy} varies non-monotonically (0.60, 0.20, 0, 0, 0.20) as n grows, while clustered C declines from 0.60 to 0 across the grid. Simultaneous arrivals (A) collapse non-monotonically to 0% at $n = 15$. Per-cell rates rest on five replicates, so per-cell Wilson 95% CIs are wide ($\approx \pm 30$–40 percentage points; e.g., the $4/5 = 0.80$ rate has CI $[0.38, 0.96]$ and the $3/5 = 0.60$ rate has CI $[0.23, 0.88]$); within-pattern rate trends should be read accordingly.

Interpretation. The absence of variation in r, r^{**}, and k across α confirms that the online arrival process, the NN tour construction, and the Core analysis all depend only on the dimensionless parameters (n, ρ) and are insensitive to the absolute scale of (L, τ). This justifies the specific choice $L = \sqrt{n}$ adopted in Section 6.1 of the main manuscript without loss of generality.

S3 Restricted Core LP and mechanism classification

This section (i) documents the restricted Core LP methodology used for scale-up instances where $2^n - 1$ enumeration is infeasible, and (ii) gives the full binding-size distributions for the 10 near-complement and 9 intermediate-coalition main-grid cases under NN.

Motivation (scale-up). The full Core LP requires one inequality per feasible coalition $S \in \mathcal{F}$, i.e., up to $2^n - 1$ constraints. For Pattern A at $n = 30$ this is $\sim 10^9$ and at $n = 50$ it is $\sim 10^{15}$, both intractable to enumerate directly.

Sampling strategy. We solve a *restricted* Core LP over a sampled subset $\mathcal{S} \subseteq \mathcal{F}$: all singletons, all feasible complements $N \setminus \{i\}$, the grand coalition, and $K = 10{,}000$ additional random intermediate-size coalitions retained after feasibility filtering. TSP costs $c^*(S)$ are computed by

Held–Karp for $|S| \leq 15$ and LKH (Helsgaun, 2000) otherwise. The first-stage nucleolus LP

$$\min_{x,\varepsilon} \ \varepsilon \quad \text{s.t.} \quad \sum_i x_i = C(N)_{\text{online}}, \qquad \sum_{i \in S} x_i - c^*(S) \leq \varepsilon \ \ (S \in \mathcal{S},\ S \neq N)$$

is solved by CBC via PuLP. If $\varepsilon^* > 0$ on the sampled subset $\mathcal{S}$, then $\varepsilon^* > 0$ on the full restricted constraint set as well (sampling drops constraints, never adds them), so Core emptiness is certified one-sidedly; the converse—concluding nonemptiness from a sample—is not valid.

Scale-up certification (Pattern A near-complement seeds). Across $n \in \{20, 30\}$ under Pattern A, the 3 cases listed below elude the main manuscript's Corollary 9 and require the restricted-LP certificate of this section:

Pattern	n	Seed	r	r^{**}	ε^*	Binding sizes	Mechanism
A	20	99	1.056	1.113	0.37	$\{n-2,\ n-1\}$	near-complement
A	20	123	1.077	1.112	0.77	$\{n-2,\ n-1\}$	near-complement
A	30	123	1.038	1.088	0.66	$\{n-2,\ n-1\}$	near-complement

The remaining seed-123 scale-up cases at $n = 50$ (and at $n \in \{20, 30, 50\}$ under Pattern C) now fire the main manuscript's Corollary 9 directly and do not require the restricted-LP certificate. The transition is the natural consequence of $r^{**} \to 1$ at the rate of the main manuscript's Theorem 16: seed 123 is not anomalous in the asymptotic limit, only uncovered by Corollary 9 at finite n below the transition. By contrast, a clean Corollary-9-fires instance at smaller scale (Pattern A, $n = 30$, seed 42) binds only at size $n - 1$.

Main-grid near-complement cases (main manuscript, Remark 12). Table S2 lists the 10 NN main-grid instances classified as near-complement by the first-stage LP's binding-size distribution.

Table S2: Ten main-grid NN instances in the near-complement class (main manuscript, Remark 12). The main manuscript's Corollary 9 and Corollary 10 are both vacuous (or the latter inapplicable). Tight binding concentrates at sizes $\{n-2, n-1\}$.

n	pattern	seed	k	r	r^{**}	r^{***}	binding at $n-2, n-1$
10	A	123	10	1.053	1.150	1.053	yes
10	B_heavy	256	10	1.126	1.138	1.061	yes
10	C	123	10	1.124	1.150	1.053	yes
15	A	42	15	1.048	1.121	1.051	yes
15	A	99	15	1.051	1.151	1.043	yes
15	A	123	15	1.075	1.145	1.042	yes
15	B_heavy	7	15	1.185	1.194	1.046	yes
15	C	42	15	1.068	1.121	1.051	yes
15	C	99	15	1.148	1.151	1.043	yes
15	C	123	15	1.141	1.145	1.042	yes

Main-grid intermediate cases (main manuscript, Observation 15). Table S3 gives the full binding-size distribution for the 9 intermediate instances. Binding concentrates on sizes $|S| \in \{3, \ldots, n-3\}$, with no size-$(n-1)$ or size-$(n-2)$ involvement.

Table S3: Nine main-grid NN instances in the intermediate-coalition class (main manuscript, Observation 15). The "binding sizes" column lists $|S| = c$ pairs, meaning c constraints of size $|S|$ are tight at the first-stage LP optimum.

n	pattern	seed	k	r	ε^*	binding sizes (count)
7	B_medium	256	5	1.436	0.028	{1=1, 2=1, 3=4, 4=6, 5=2}
10	B_medium	123	8	1.423	0.192	{1=1, 2=3, 3=6, 4=9, 5=9, 6=5, 7=1}
10	B_medium	256	8	1.358	0.217	{1=2, 2=2, 3=1, 4=2, 5=1, 6=3, 7=3, 8=1}
10	E	42	8	1.495	0.062	{1=1, 2=1, 3=5, 4=3, 5=3, 6=2, 7=1}
12	B_medium	99	10	1.395	0.744	{3=2, 4=2, 5=6, 6=12, 7=12, 8=6, 9=1}
12	B_medium	123	10	1.487	0.767	{1=1, 2=1, 3=2, 4=2, 5=6, 6=9, 7=7, 8=4, 9=1}
12	B_medium	256	10	1.423	0.941	{1=1, 2=1, 6=3, 7=9, 8=9, 9=5, 10=1}
12	E	42	9	1.610	0.289	{3=4, 4=5, 5=5, 6=5, 7=3, 8=1}
15	B_medium	256	12	1.326	0.185	{1=1, 2=1, 3=3, 4=2, 7=1, 8=5, 9=7, 10=7, 11=2, 12=1}

Interpretation. The near-complement table (Table S2) confirms that the 10 main manuscript's Remark 12 cases share the signature $\{n-2, n-1\}$ binding: neither size alone certifies emptiness, but the combined presence of both sizes does. The intermediate table (Table S3) shows that the 9 Observation 15 cases have binding concentrated on small-to-medium coalition sizes, with $r \in [1.33, 1.61]$ and $k < n-1$ in every case. A tight analytic sufficient condition for either class remains open (main manuscript, Section 7).

Solver and calibration. For $n \leq 20$ we use exact Held–Karp (pure distance); for $n > 20$ we use the LKH heuristic of Helsgaun (2000) via the `elkai` Python wrapper. Calibration against Held–Karp on 15 instances at $n = 15$ yielded a relative error of 0.000% in every case, so at this scale LKH is effectively exact for our pure-distance $c^*(S)$. Where the full Core LP is intractable to enumerate (pattern A at $n \geq 30$), we use a *restricted* Core LP over a sample $\mathcal{S} \subseteq \mathcal{F}$ consisting of all singletons, all feasible complements $N \setminus \{i\}$, the grand coalition, and 10,000 randomly sampled intermediate-size coalitions. A strictly positive first-stage LP optimum certifies Core emptiness on the sample, a one-sided certificate for the full Temporal Core (it cannot conclude nonemptiness). Full methodology and the near-complement / intermediate-coalition case tables are in Supplementary Materials §S3.

Temporal Nucleolus cascade implementation. The Core-stability results of the main manuscript (Sections 5 and 6) depend only on the *first-stage* LP of the cascade (equation (1) of the main manuscript): the Temporal Core is nonempty iff $\varepsilon_1^* \leq 0$, and the sign is independent of tie-breaking. For Core-existence judgment—which is the focus of the main manuscript—the first-stage LP alone therefore suffices. For the full Temporal Nucleolus point, the sequential cascade of Guajardo & Jörnsten (2015) is applied. Our implementation adopts a simplified variant that fixes all tight constraints at each stage, sufficient under primal non-degeneracy; handling primal degeneracy via rank-aware tight-constraint selection, due to **?** and clarified by Guajardo & Jörnsten (2015), is a straightforward extension and does not affect any of the Core stability results.

S4 Balanced-near-complement: coverage audit

This section documents the empirical coverage of Corollary 13 (balanced-near-complement threshold) of the main manuscript on the 10 NN main-grid near-complement cases of Remark 12. The proof of Corollary 13 is given in the main manuscript; the diagnostic material below supplements the coverage statistics summarized there.

Balancing LP. Let

$$\begin{aligned} B_{n-1} &= \{N \setminus \{i\} : i \in N\}, \\ B_{n-2} &= \{N \setminus \{i,j\} : i \neq j \in N\}, \end{aligned}$$

and, for an instance with $\mathcal{F} \supseteq B_{n-1} \cup B_{n-2}$, let

$$\begin{aligned} r^{(\diamond)}_{\min} \;:=\; \min\Big\{ & r^{(\diamond)}(\lambda) : \lambda \geq 0, \\ & \sum_{S \ni i} \lambda_S = 1 \ \forall i \in N, \\ & \operatorname{supp}(\lambda) \subseteq B_{n-1} \cup B_{n-2} \Big\} \end{aligned}$$

denote the sharpest threshold in the proposition's family. By Corollary 13, $r > r^{(\diamond)}_{\min}$ implies $\mathrm{Core}(N, c, \mathcal{F}) = \emptyset$. The LP is solved with $n + \binom{n}{2}$ non-negative variables and n equality constraints; the reference implementation is `code/scripts/near_complement_coverage_check.py` and its output is `code/results/near_complement_coverage_check.csv`.

Coverage on the 10 near-complement cases. For every one of the 10 NN main-grid near-complement instances (Table S2), we have $|B_{n-1} \cap \mathcal{F}| = n$ and $|B_{n-2} \cap \mathcal{F}| = \binom{n}{2}$ (all 10 cases arise at $k = n$, so every S of size $n-1$ or $n-2$ is feasible). The proposition's feasibility hypothesis is therefore met on all 10; the empirical question is only whether the threshold inequality $r > r^{(\diamond)}_{\min}$ fires.

The proposition fires on 9 of 10 cases. The single residual ($n = 15$, Pattern A, seed 42) has $r = 1.0475 < r^{(\diamond)}_{\min} = 1.0495$ by a margin of $\approx 2 \times 10^{-3}$.

Diagnostic for seed 42. The proposition's feasibility hypothesis is fully met, so the proposition is not inapplicable; the mixed $B_{n-1} \cup B_{n-2}$ collection is simply not rich enough to certify this particular instance's emptiness. The first-stage Core LP on the full $\mathcal{F}$ still certifies the Core empty (the instance is classified near-complement in the CSV), with binding concentrated at sizes $n - 1$

Table S4: Balanced-near-complement coverage audit (main manuscript, Corollary 13). Fires iff $r > r_{\min}^{(\diamond)} + 10^{-9}$.

n	pattern	seed	r	$r_{\min}^{(\diamond)}$	$r - r_{\min}^{(\diamond)}$	fires
10	A	123	1.0529	1.0490	+0.0039	yes
10	B_heavy	256	1.1261	1.0480	+0.0781	yes
10	C	123	1.1238	1.0490	+0.0748	yes
15	A	42	1.0475	1.0495	−0.0020	**no**
15	A	99	1.0507	1.0402	+0.0105	yes
15	A	123	1.0749	1.0414	+0.0335	yes
15	B_heavy	7	1.1849	1.0421	+0.1428	yes
15	C	42	1.0680	1.0495	+0.0185	yes
15	C	99	1.1476	1.0402	+0.1074	yes
15	C	123	1.1406	1.0414	+0.0992	yes

and $n-2$ simultaneously; but the binding geometry cannot be captured by a non-negative balancing over exactly those two sizes without violating the efficiency equality by the observed margin.

Refined open problem. Whether a strict extension of the balancing family closes the seed-42 residual is left open. Two natural enlargements suggest themselves: (i) add size-$(n-3)$ coalitions, i.e., enlarge the support to $B_{n-1} \cup B_{n-2} \cup B_{n-3}$ and recompute $r_{\min}^{(\diamond)}$ (this requires $\binom{15}{3} = 455$ additional size-$(n-3)$ coalition costs per instance); (ii) replace the balancing equality $\sum_{S \ni i} \lambda_S = 1$ by the covering inequality $\sum_{S \ni i} \lambda_S \geq 1$, which preserves the Bondareva–Shapley argument by a standard slack-variable construction. We tested enlargement (i) directly: solving the LP over $B_{n-1} \cup B_{n-2} \cup B_{n-3}$ on the seed-42 instance yields $r_{\min}^{(\diamond,3)} = 1.047878$ versus the realized $r = 1.047533$, narrowing the negative margin from -1.92×10^{-3} to -3.45×10^{-4} (a 5.6× tightening) but still failing the strict inequality; on the other 9 cases the enlargement leaves $r_{\min}^{(\diamond)}$ either unchanged or only marginally tighter, without altering the fires/no-fires classification (reference implementation `code/scripts/near_complement_coverage_check_k3.py`, output `code/results/near_complement_coverage_check_k3.csv`). The residual at seed 42 therefore persists under this single enlargement step; whether the covering-inequality variant (ii), a further B_{n-4} enlargement, or some non-balancing certificate closes the gap remains open and is deferred to future work.

No-false-positives note. Corollary 13 is a proved sufficient condition; any instance on which the balancing LP fires is guaranteed to have an empty Core. We therefore do not need a separate false-positive sweep over the 108 observed nonempty-Core main-grid instances; the certification is unilateral by construction.

S5 Robustness to dispatch policy

We rerun the 175-instance grid under three dispatch policies: nearest-neighbor (NN, the baseline used in the preceding subsections), cheapest-insertion (CI), and batch re-optimization (BR, which

Table S5: Dispatch-policy robustness over the 175-instance grid. $\bar{r}$ is the mean competitive ratio; the last columns report how often each analytic mechanism fires and how many intermediate-coalition empty-Core cases (Observation 15) arise.

Policy	$\bar{r}$	Core rate	Empty	Cor 9 app/fires	Cor 10 app/fires	Obs 15
Nearest-neighbor	1.312	0.617	67	96/37	80/57	9
Cheapest-insertion	1.223	0.823	31	97/17	80/30	0
Batch re-optimization	1.201	0.857	25	97/ 7	80/25	0

solves the shortest Hamiltonian path from the vehicle's current position through the waiting queue U_t back to the depot by Held–Karp at each dispatch event). Table S5 summarizes the aggregate outcomes on all $175 \times 3 = 525$ policy-instance pairs.

Three observations stand out. First, the mean competitive ratio decreases monotonically from 1.312 (NN) to 1.223 (CI) to 1.201 (BR), reflecting the increasing quality of the online plan. Second, the empirical Core-existence rate increases correspondingly: NN yields 67 empty Cores (38%), CI 31 (18%), and BR 25 (14%). Third, and central to our theoretical claims, *every* instance in which the Corollary 9 hypothesis $r > r^{**}$ holds is observed empty (37/37 for NN, 17/17 for CI, 7/7 for BR), and *every* instance in which Corollary 10's hypothesis $r > r^{***}$ holds is likewise observed empty (57/57, 30/30, 25/25), with no false positives under any policy. The Corollary 19 safeguard holds analytically without exception across all three policies: no instance with $k < n-1$ has $r > r^{**}$ or $r > r^{***}$. Intermediate-coalition empty Cores (Observation 15) appear exclusively under NN; CI and BR produce *zero* intermediate cases in the 175-instance grid, corroborating the policy-sensitivity of that mechanism.

Corollary 9 is applicable in 96–97 of the 175 instances and Corollary 10 in 80, with applicability counts differing by at most one across NN/CI/BR. The thresholds r^{**} and r^{***} depend only on TSP costs and the arrival sequence (policy-independent); their applicability (whether $N \setminus \{i\} \in \mathcal{F}$) depends on service-time windows (weakly policy-dependent), and only the realized r varies meaningfully with policy — the source of the monotone drop in fires counts across NN/CI/BR in Table S5.

S6 Illustrative example and reduction at simultaneous arrivals

S6.1 Example: Solomon C101 first five customers

Example 1 (Solomon C101, first five customers)**.** We illustrate the Online TSG on the first five customers of the Solomon C101 benchmark (Solomon, 1987). Depot is at $(35, 35)$ and coordinates/ready times are as in Table S6. The online policy is nearest-neighbor departing from the depot at $t = 0$.

Euclidean distances give $c^*(N) = 77.593$ and realized online cost $C(N)_{\text{online}} = 80.059$, so $r = 1.032$. Customers 1–4 have $s_i = a_i$ because the vehicle reaches each of them before its ready window opens; under Definition 1's closed-interval convention their singleton windows $\mathrm{CW}(\{i\}) = [a_i, a_i] =$

Table S6: Solomon C101 first five customers used in Example 1.

i	(x_i, y_i)	ready a_i	due	$c(\{i\})$	service s_i
1	(45, 68)	912	967	68.964	912.00
2	(45, 70)	825	870	72.801	825.00
3	(42, 66)	65	146	63.561	65.00
4	(42, 68)	727	782	67.469	727.00
5	(40, 69)	15	67	68.731	34.37

$\{a_i\}$ are nonempty, so every $\{i\}$ lies in $\mathcal{F}$ automatically. Because the vehicle spends most of its time idle waiting for each successive ready window, the service timeline is fully sequential in the sense of Lemma 5: one verifies that $s_j \le a_{j+1}$ for every consecutive pair in the ready-time order $(5, 3, 4, 2, 1)$. Consequently, by Corollary 11 (equivalently Corollary 19, since $k = \max_t |U_t| = 1 < n - 1 = 4$), Corollary 9 is vacuous: *no* complement coalition $N \setminus \{i\}$ is feasible, so r^{**} is undefined here because the minimum in its definition is taken over the empty set. The inequality $r > r^{**}$ is thus not testable and Corollary 9 makes no prediction on this instance.

Using $\sum_i x_i = C(N)_{\text{online}}$ as the efficiency budget, the Temporal Nucleolus minimizes $\max_i\{x_i - c(\{i\})\}$ subject to efficiency, yielding the equal-excess allocation

$$
\begin{aligned}
&(x_1, x_2, x_3, x_4, x_5) \\
&\quad = (16.67,\ 20.51,\ 11.27,\ 15.18,\ 16.44),
\end{aligned}
$$

with uniform slack $c(\{i\}) - x_i = 52.29$ across customers. (The displayed coordinates are rounded to two decimals; the efficiency equality $\sum_i x_i = C(N)_{\text{online}} = 80.059$ holds exactly at the unrounded LP optimum.) The static TSG allocation (computed on $2^N \setminus \{\emptyset\}$, treating every coalition as feasible) is instead the Nucleolus of the full 31-constraint game; this example shows how temporal restriction discards 26 of those 31 constraints while preserving core existence.

S6.2 Reduction at simultaneous arrivals

Proposition 2 (Reduction at simultaneous arrivals)**.** *Suppose $a_1 = a_2 = \cdots = a_n$ and $s_i > a_1$ for every $i \in N$, so all customers share a common waiting window. Then $\mathcal{F} = 2^N \setminus \{\emptyset\}$, and the Temporal Nucleolus of $(N, c, \mathcal{F})$ equals the lexicographic minimizer of excess on the static TSG constraint family over proper coalitions $S \subsetneq N$ with efficiency budget $C(N)_{online}$. When the realized tour is additionally optimal (i.e. $C(N)_{online} = c^*(N)$), this coincides with the classical static Nucleolus of Potters et al. (1992).*

Proof. By assumption there exists $t^\star \in [a_1, \min_i s_i)$ with $U_{t^\star} = N$. Every $S \subseteq N$ satisfies $S \subseteq U_{t^\star}$, so $\mathrm{CW}(S) \supseteq [a_1, \min_i s_i) \neq \emptyset$ and $\mathcal{F} = 2^N \setminus \{\emptyset\}$. The Temporal Core (Definition 3) imposes constraints for every proper subset $S \subsetneq N$—i.e. the full static TSG inequality family—together with the efficiency equality $\sum_i x_i = C(N)_{\text{online}}$. The sequential-LP cascade (1) thus coincides with Kohlberg's procedure for the static TSG on this constraint family, and the lexicographic minimizer is unique. When $C(N)_{\text{online}} = c^*(N) = c(N)$, the efficiency budget matches the classical static-Nucleolus LPs and the allocation recovered is that of Potters et al. (1992). In general, the Temporal Nucleolus under simultaneous arrivals is the lexicographic minimizer on the same constraint family but with a shifted efficiency budget. $\square$

Numerical verification. For the 25 pattern-A (simultaneous) instances of the main-grid experimental design, we compared the Temporal Nucleolus with the static Nucleolus computed on the unrestricted coalition family $2^N \setminus \{\emptyset\}$. The two coincide coordinate-wise in every instance (maximum absolute deviation $< 10^{-9}$), confirming Proposition 2.

S7 Inheritance of static Core stability

Proposition 3 (Inheritance of Static Core stability)**.** *If $C(N)_{online} = c^*(N)$, i.e. the online tour is optimal, then every static-TSG Core allocation remains in the Temporal Core; in particular* $\mathrm{Core}_{static}(N, c) \subseteq \mathrm{Core}(N, c, \mathcal{F})$.

Proof. Let $x \in \mathrm{Core}_{\text{static}}$. Efficiency is inherited since $\sum_i x_i = c(N) = c^*(N) = C(N)_{\text{online}}$. For any $S \in \mathcal{F}^{\mathrm{p}} \subseteq 2^N \setminus \{\emptyset, N\}$, the static constraint $\sum_{i \in S} x_i \leq c(S)$ is already imposed on x, so it also holds in the temporal setting. $\square$

The proposition is a static-equality consequence: when the online dispatch policy realizes the offline-optimal tour, no extra efficiency budget is opened up by online overhead, and the static Core's coalition-level inequalities transfer verbatim to the smaller temporal constraint family $\mathcal{F}^{\mathrm{p}} \subseteq 2^N \setminus \{\emptyset, N\}$. The proposition is therefore informationally weakest in the simultaneous-arrival regime ($\mathcal{F} = 2^N \setminus \{\emptyset\}$) where it coincides with the static Core itself, and operationally vacuous in the typical online case ($C(N)_{\text{online}} > c^*(N)$). The structural-survival counterpart with bite—the obstruction at $k < n-1$ that blocks all complement-based mechanisms—is Corollary 19 of the main manuscript.

S8 Auxiliary feasibility lemmas

The two lemmas below characterize feasibility of complement coalitions $N \setminus \{i\}$ in $\mathcal{F}$ under the realized arrival-service sequence, and record a structural consequence of fully sequential arrivals. They are used in the main manuscript inside Corollary 9's discussion and Corollary 11's proof.

Lemma 4 (Feasibility of $N \setminus \{i\}$)**.** *For each $i \in N$, $N \setminus \{i\} \in \mathcal{F}$ if and only if $\max_{j \neq i} a_j < \min_{j \neq i} s_j$.*

Proof. ($\Rightarrow$) If $N \setminus \{i\} \in \mathcal{F}$ there is t with $N \setminus \{i\} \subseteq U_t$. Then $a_j \leq t < s_j$ for every $j \neq i$, so $\max_j a_j \leq t < \min_j s_j$ (both over $j \neq i$). ($\Leftarrow$) Conversely, if $\max_{j \neq i} a_j < \min_{j \neq i} s_j$, any t in the open interval between them satisfies $a_j \leq t < s_j$ for all $j \neq i$, so $N \setminus \{i\} \subseteq U_t$ and $N \setminus \{i\} \in \mathcal{F}$. $\square$

Lemma 5 (Fully sequential arrivals preclude $N \setminus \{i\} \in \mathcal{F}$)**.** *If arrivals are fully sequential in the sense that $s_j \leq a_{j+1}$ for all $j = 1, \ldots, n-1$ (each customer is served before the next arrives), then $N \setminus \{i\} \notin \mathcal{F}$ for every $i \in N$.*

Proof. Fix i and assume customers are indexed in arrival order $1, \ldots, n$ so that $a_1 \leq \cdots \leq a_n$. Let $\ell^\star = \max\{j : j \neq i\}$ and $j^\star = \min\{j : j \neq i\}$ denote, respectively, the latest and earliest arrival-order indices other than i. By the sequential-arrival hypothesis $s_j \leq a_{j+1}$ applied along the chain $j^\star, j^\star + 1, \ldots, \ell^\star - 1$, we obtain $s_{j^\star} \leq a_{j^\star+1} \leq s_{j^\star+1} \leq \cdots \leq a_{\ell^\star}$, hence $a_{\ell^\star} \geq s_{j^\star}$. Therefore $\max_{j \neq i} a_j \geq \min_{j \neq i} s_j$, violating the if-and-only-if criterion of Lemma 4. □

S9 Partition-pair certification of the 9 intermediate cases

This section reports, for each of the 9 intermediate-coalition empty-Core instances of Observation 15, the partition pair $(S, N \setminus S)$ supporting the dual of the first-stage Core LP, together with the corresponding partition-pair threshold $r_S^{(P)}$ of Theorem 14. The partition pairs were obtained by re-solving the first-stage LP and extracting nonzero dual prices; the reference implementation is `code/scripts/intermediate_dual_analysis.py` and its output is `code/results/intermediate_dual_analysis` / `partition_pair_threshold.json`.

Table S7: Partition-pair certificates for the 9 intermediate-coalition cases under nearest-neighbor dispatch (Observation 15). For each instance, the first-stage Core LP's dual concentrates on a single partition pair $(S, N \setminus S)$ with $|S|, |N \setminus S| \geq 2$; both halves are feasible. Theorem 14 fires whenever $r > r_S^{(P)} := [c(S) + c(N \setminus S)]/c^*(N)$, with margin reported in the last column. All 9 fire; the structural obstruction of Corollary 19 blocks every *singleton*-complement mechanism but not the partition-pair mechanism via non-singleton splits.

n	pattern	seed	$\lvert S\rvert$	$\lvert N \setminus S\rvert$	$c(S)$	$c(N \setminus S)$	$r_S^{(P)}$	$r - r_S^{(P)}$
7	B_medium	256	2	5	3.111	6.232	1.428	+0.008
10	B_medium	123	5	5	7.888	5.887	1.384	+0.039
10	B_medium	256	8	2	9.028	3.719	1.313	+0.045
10	E	42	3	7	7.270	8.365	1.484	+0.012
12	B_medium	99	9	3	10.654	6.529	1.284	+0.111
12	B_medium	123	3	9	6.095	9.543	1.354	+0.133
12	B_medium	256	10	2	11.610	4.074	1.271	+0.152
12	E	42	4	8	8.036	10.391	1.561	+0.049
15	B_medium	256	12	3	13.407	5.167	1.300	+0.026

No-false-positives. Theorem 14 is a proved sufficient condition (Bondareva–Shapley applied to the balanced family $\{S, N \setminus S\}$ with weights 1 each), so every instance where it fires has an empty Core by construction; no false-positive sweep is required.

Reading. The signature $|S| + |N \setminus S| = n$ is the defining property of a partition pair, and is always present when the dual support has exactly two coalitions. The minimum-size half ranges from $|S| = 2$ to $|S| = 5$ across the 9 cases; the partition pair is therefore typically asymmetric, which is consistent with the realized arrival timing creating a bottleneck between an early-arriving cluster and a late-arriving cluster.

S10 Threshold-distribution histograms

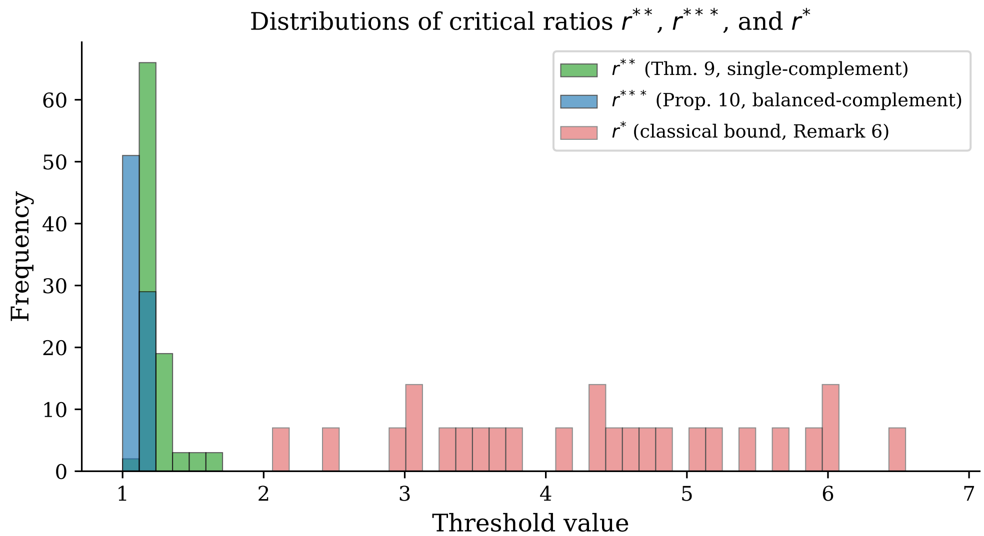

Figure S2: Histograms of r^{**}, r^{***}, and r^{*} over all NN instances. r^{**} and r^{***} concentrate near the empirical r (making them informative Core-emptiness thresholds via Corollary 9 and Corollary 10); r^{*} lies far to the right and rarely binds.